

Class 30006 – Financial Markets and Institutions
Università Commerciale Luigi Bocconi
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Are Financial Markets Efficient? **(Chapter 6)**



Outline

On the principles of market efficiency

- 1) arbitrage and short-selling
- 2) the efficient market hypothesis
- 3) look at evidence for, and against, the efficient market hypothesis
- 4) a quick look at *behavioral finance*

Motivation

So far in this class, we have often mentioned **expectations**:

1. Expectations of future inflation and interest rates
 - important for: bond's equilibrium interest rate
2. Future expected earnings/dividends
 - important for: stock prices
3. (Expected) spot prices “in the future”
 - Important for: derivatives payoff

But we have never said anything about how these expectations are formed

Motivation

- ✓ We shall say that investors are **rational**
 - what does that mean?
 - that they form their “best” guess of the future using **all available information**
- ✓ We do not need *everyone* to be rational
 - but *if a few (“smart money”) are*, then this is sufficient to guarantee that ...
... **market prices reflect “all” available information**

This, in essence, is the
Efficient Market Hypothesis (EMH)

The rationale behind it is

ARBITRAGE

Arbitrage (1/4)

- ✓ We have encountered **arbitrage** when we talked about futures converging to spot prices at delivery
- ✓ EX: if the future price at delivery date is higher than the spot price ($F_T > S_T$), you can buy at spot and sell a future. Earn risk-free profit. Reverse if ($F_T < S_T$)

Arbitrage opportunities

... are opportunities of making a “*risk-free*” profit by exploiting mis-pricings in the market*

*: academic definition, in reality always run some risk.

Arbitrage (2/4)

✓ Other arbitrage examples

- a) same asset trades with different prices on two different exchanges
- b) two assets with identical cash flows & risk but different prices
- c) an asset with a known price in the future does not trade today at the price discounted by the risk-free interest rate

✓ What happens when arbitrage opportunities are being exploited? They **disappear**! ... because of market forces

Indeed:

- a) buy cheaper asset $\Rightarrow$ demand $\uparrow \Rightarrow$ price of cheaper asset $\uparrow$
- b) sell more expensive asset $\Rightarrow$ supply $\uparrow \Rightarrow$ price of expensive assets $\downarrow$
- c) prices change because of law of demand
 - when supply and demand converge $\Rightarrow$ no more arbitrage opportunities!

Arbitrage (3/4)

Short selling and arbitrage

EX: you know with certainty (private information) that an asset is overpriced, and price will fall in the future

- ✓ Strategy: sell something *you don't have*, *betting* on expected price change
 1. you borrow the asset from someone for a rental fee at $t=0$
 2. sell it at $t=0$ (say S_0)
 3. buy it again in the future ($t=1$) when it has a lower price (say: $S_1 < S_0$: *this is your bet!*)
 4. return the asset ($t=1$) to its rightful owner and pay rental fee!
 - profit = $S_0 - S_1 - \text{rental fee}$
- ✓ *Arbitrage by short-selling has corrective effect on the price*
 - selling of the asset by short-sellers reduces its price
 - soon enough, the arbitrage opportunity will have been eliminated

Arbitrage (4/4)

✓ *Short-selling is also used for risky speculation*

- quite risky w.r.t to buying a risky asset and resting on its limited liability
- essentially your bet on a price change in both cases, but with short selling. earn/lose a lot: quite risky! Indeed:
 - potential **gain**: limited to the spot price of the asset (in case the asset becomes entirely worthless: $S=0$)
 - potential **loss** unlimited: you have to buy back and return the asset, no matter what it costs at that point ($S_1 \gg S_0$!)

Arbitrage and EMH

- ✓ Arbitrage opportunities cannot survive for long in a competitive market
 - there are other investors who will see a profit and take advantage of it
 - if you found one opportunity: very lucky!

- ✓ The fact that there are few arbitrage opportunities...
 - ... tells you that there are few securities whose price does not reflect all publicly available info (otherwise there would be an easy arbitrage opportunity)

This is the Efficient Market Hypothesis (EMH)

EMH definition

Definition of **EMH** (semi-strong form)

- ✓ “The security’s price reflects **all available public information** at all times”
- ✓ this hypothesis implies that the price of a stock in the future is not predictable from any information that is public knowledge (else, there would be arbitrage opportunities)
- ✓ it does **NOT** say that you cannot profit from private information: that is not incorporated in the stock price

EMH definition (1/2)

There also 2 other definitions of EMH

- ✓ **Strong efficiency:** all information, whether *public* or *private*, is priced-in. Prices reflect the true, fundamental (intrinsic) value of securities (= all items that have direct impact on future income streams of the security)

$$P = \text{fundamental} = \sum_{t=1}^{\infty} \frac{Div_t}{(1 + k_e)^t}$$

- *much less popular today because ...*
- *... it rules out the existence of **speculative bubbles**. Have bubbles when there is detachment from fundamentals*
 - *ex.: people buy because they expect the price to go up, or just because see others buy it, not because of any idea that fundamentals have improved*

EMH definition (2/2)

- ✓ **Weak efficiency:** all past stock prices are reflected in today's stock price
 - *current prices fully reflect all information contained in past prices*
 - *this form implies that past prices cannot be used to predict future stock price movements*
 - *therefore, it is not possible for a trader to make abnormal returns by using only the past history of prices ...*
 - *in other terms, technical analysis is pointless*
 - *can replace investment professionals with a monkey to predict future stock prices!*



Evidence on Efficient Market Hypothesis

Favorable Evidence

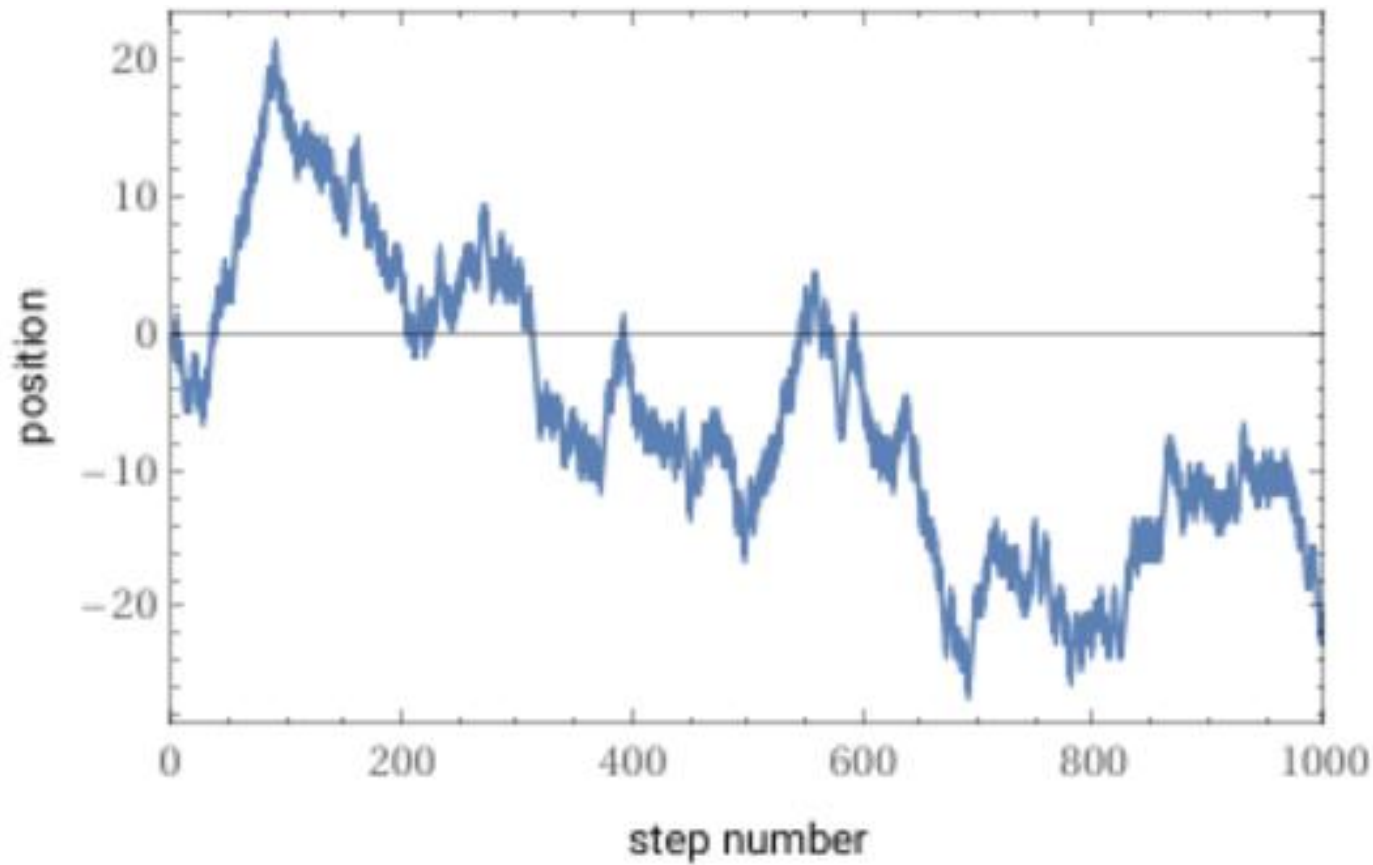
1. **Random Walk** behavior of stock prices
2. **Anticipated announcements** don't affect stock price
3. **You can't beat the market** (not even analysts and funds)
4. **On the book** (more examples)

1 Random Walk of Stock Prices

- ✓ If EMH is right, prices should be a random walk
- ✓ That is, **future changes in stock prices should be unpredictable**
- ✓ Random walk technically means that the best predictor for tomorrow's price is today's price
 - why is this “random”? Because if today is your best guess for tomorrow then you really have no idea about tomorrow
 - $P_{t+1} = P_t + e_t$ where: e_t is random error with zero mean and finite variance, it reflects the new information (unpredictable) arriving between $t+1$ and t .
 - $\Delta P_{t+1} = P_{t+1} - P_t = e_t$ In other words, changes in stock prices have the same distribution, are independent of each other (4 ex.: ΔP_{t+1} is independent of past price changes, say ΔP_t or ΔP_{t-2} , etc...)
- ✓ Many studies find evidence on random walk behavior of stocks using stock market data

1 Random Walk of Stock Prices

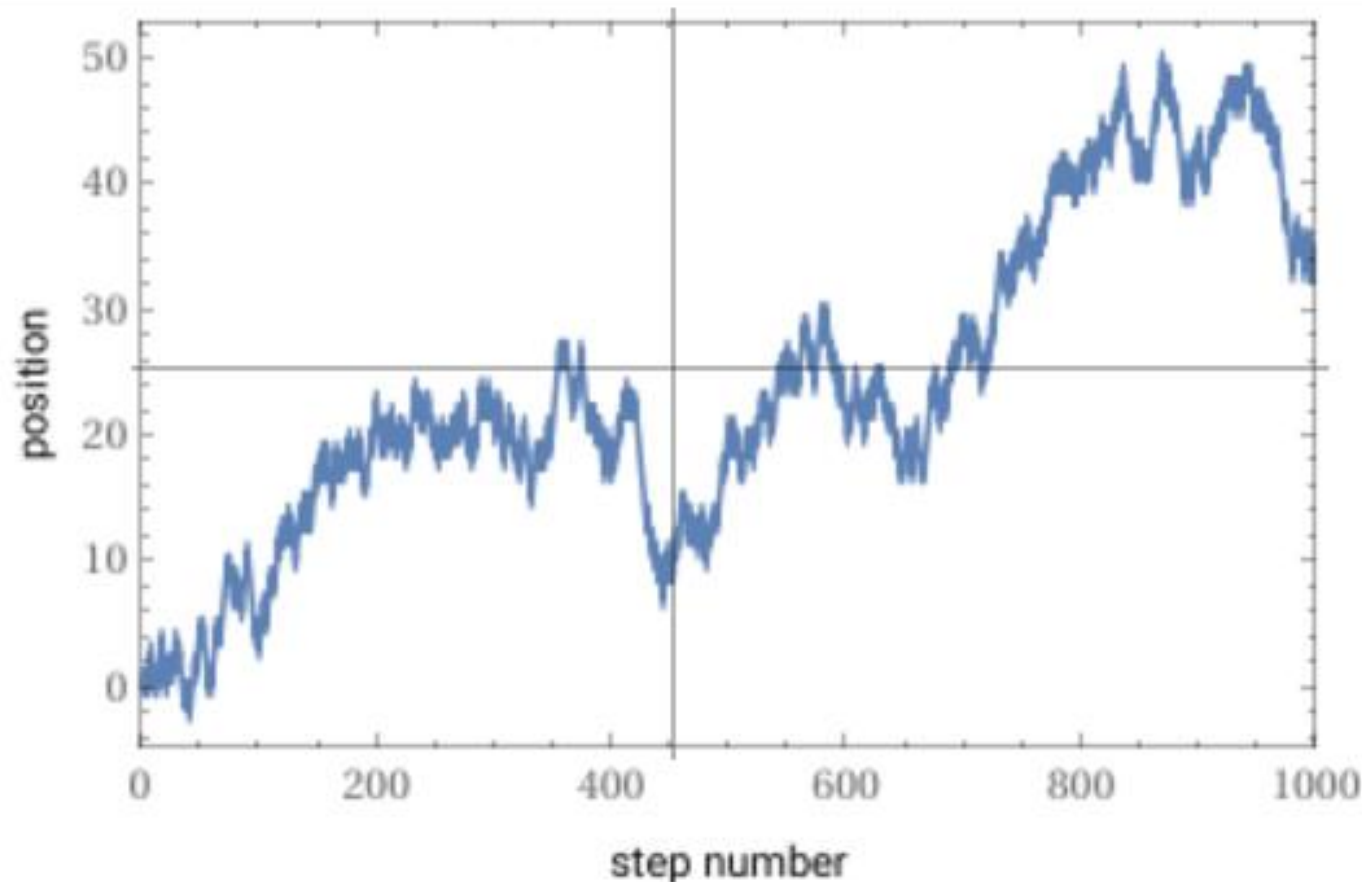
- to make a random walk pattern click [here](#)!*



*: for fun, try also [this](#) to control for drift!

1 Random Walk of Stock Prices

- click [again](#) (on “New Sample”) ... you get different graph!
- you can produce all kinds of patterns with this!
- thus, many things that look “systematic” to the human eye, may just be randomness 😊!



2 Un/Anticipated Announcements

- ✓ A positive (negative) but **anticipated** announcement about a company* will not affect (raise or decrease) the price of its stock
- ✓ Why? because this information was already reflected in the stock price (it was anticipated ... already priced in)
- ✓ **Un**anticipated announcements will affect the stock price, but there should be a full immediate adjustment to the new price under EMH

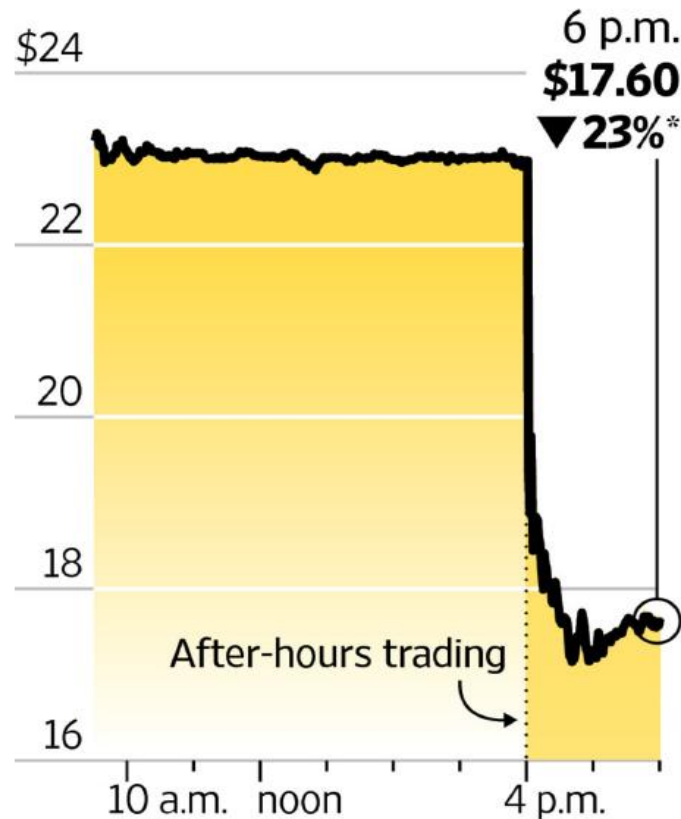
*: note this is *public* information

Snapchat Parent Posts Disappointing User Growth; Stock Plunges

Rival Facebook has been nibbling away at features that made Snapchat stand apart

Aw Snap

Snap shares, Wednesday



Source: [WSJ, May 10, 2017](#)

Can't beat the market

- ✓ If EMH is right, investment analysts and mutual funds should not be able to consistently **beat the market**
- ✓ It means that buying the shares they suggest should not be better than just buying a basket of the market as whole

EMH Implications

EMH does **not** imply that:

1. ... prices **do not change**:
 - as new information comes in, securities are re-evaluated
2. ... a **rational** investor always gets the price right:
 - only that his/her **expectations** are correct (“on average” he/she is right)
3. ... a zero average market returns:
 - earn a “fair” return in equilibrium, based on the riskiness of the security

Exercise: EMH

- ✓ A firm unexpectedly announces today that it will increase its dividend by 20% starting next year
- ✓ The price of the stock today does not change
- ✓ Exactly one year from today, just before the announced dividend increase, the stock price goes up by 20%

Q: is this pattern of price movements consistent with EMH?

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Exercise: EMH

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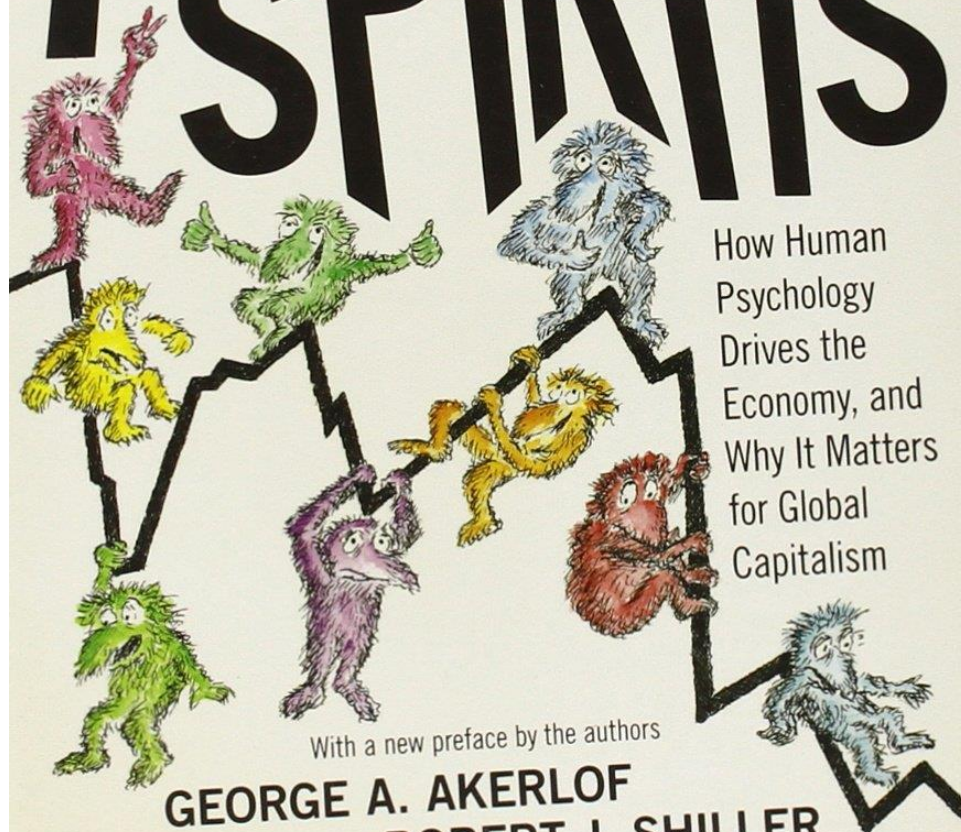
A: NO! The price should rise today because new public info is immediately priced in

EMH: does it really work?

- ✓ All the evidence in favor of EMH led its father and most prominent advocate, *Eugene Fama*, the 2013 Nobel prize in Economics, to declare in 1970:
- ✓ *“The evidence in support of EMH is extensive and, somewhat unique in economics, contradictory evidence is sparse”*
- ✓ This view is not shared by another 2013 Nobel prize winner, *Robert Shiller*
 - he wrote a book called “Irrational Exuberance” and one called “Animal Spirits”

WINNERS OF THE NOBEL PRIZE IN ECONOMICS

ANIMAL SPIRITS



How Human
Psychology
Drives the
Economy, and
Why It Matters
for Global
Capitalism

With a new preface by the authors

GEORGE A. AKERLOF
and **ROBERT J. SHILLER**

Evidence Against EMH

Unfavorable Evidence

- A. **January effect**
- B. **Small-firm effect**
- C. **Market overreaction and momentum**
- D. **Excessive volatility**

None of these phenomena can be fully explained in a framework in which the efficient market hypothesis is valid

Evidence Against EMH

A. January effect

- ✓ Stock prices fall in December and rise in January
 - **Abnormal positive return** in the month of January that is predictable and, hence, inconsistent with random-walk behavior

Evidence Against EMH

B. Small-firm effect

- ✓ Many empirical studies have shown that **small firms** have earned **abnormally high returns** over long periods of time, even when the greater risk for these firms has been considered
 - this is a challenge for EMH because no stock should consistently exhibit excess returns

Evidence Against EMH

C. Market overreaction and momentum:

- ✓ sometimes stock prices **overreact to news announcements** (*overshooting*)
- ✓ sometimes prices *continue to rise* for quite some time after a positive news (**momentum**)
- ✓ Momentum strategies: sell stocks that have had bad returns and buy those with high returns in the last 3-12 months



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Oil price jumps to \$89 a barrel following Hamas's attack on Israel

Conflict fuels concerns of wider uncertainty in the Middle East

October 7, 2023
Hamas attack!
Saturday (markets
are closed)

October 9, 2023
(market opening)

Look at immediate
adjustment!

But later ...





Longer term view (after October 7, 2023):

- oil price decreased
- it took a downward trend

Oct 2025 (after peace agreements):

- only limited downward adjustment of oil prices



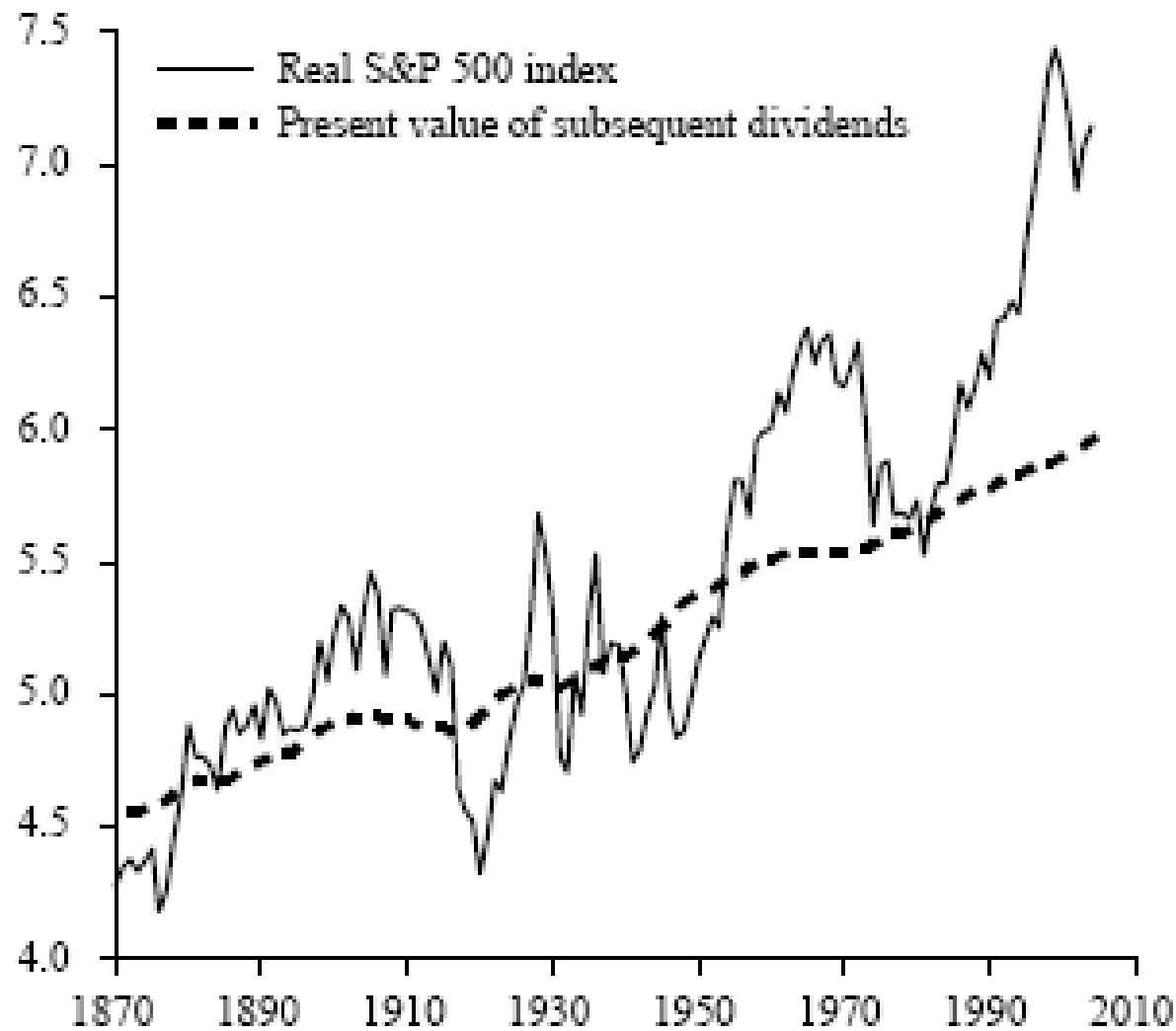
Evidence Against EMH

D. Excessive volatility:

- ✓ EMH implies that prices basically reflect all and only **market fundamentals** (PV of future dividends)
- ✓ However, stocks appear to have **excessive volatility** compared to volatility of fundamentals
- ✓ 4 ex.: Shiller found that fluctuations in the S&P 500 could not be justified by the subsequent fluctuations in the dividends of the stocks making up this index
- ✓ Shiller concluded that there must be factors other than fundamentals affecting stock prices

Figure 1
Real S&P 500 index and present value of subsequent
dividends

Real price (log scale)



Behavioral Finance

- ✓ New field that includes concepts from psychology, sociology, and other social sciences to understand investors and patterns of securities prices
 - Shiller seminal works, Richard Thaler won the 2017 Nobel prize
- ✓ *Loss aversion*
 - loss aversion (*feel worse when you lose than better when you gain*) may explain why so few people ever go short on a stock: normally when going short (borrow to sell), you first see losses on your position which doesn't make you feel good about it
- ✓ *Overconfidence*
 - people tend to be overconfident in their own decisions (almost everybody thinks of himself as an above-average investor). Investors overconfidence may perpetuate stock price bubbles
- ✓ *Herd behavior*
 - our decisions are influenced by the decisions of others around us; give up private information to follow the herd ...

Behavioral Finance: Evidence

- ✓ La Porta (1996): expectations of **stock market analysts** have predictive power for future stock returns
- ✓ Result: companies whose analysts are **most optimistic** about earn **poor returns** relative to alternative companies
- ✓ Interpretation:
 - analysts (and investors who follow them) are **too optimistic** about stocks with **rapidly growing earnings**
 - they **overestimate** the probability that the firm is the **next Google** based on recent data (but “real” Googles are actually very rare)

Outside of Academia

- ✓ Warren Buffett does not believe in EMH. Why?
 - because he proved he can beat the market! 19.7% per year for the last 49 years (9.8% for S&P)
- ✓ He outlined his ideas in a famous 1984 lecture, “[The Superinvestors of Graham-and-Doddsville](#)” in honor of two Columbia Professors, Graham and Dodd, who taught to Buffett and many other famous investors
- ✓ Buffett and other Graham and Dodd’s students, promote the idea of **value investing**:
 - going long in underpriced companies on the basis of their fundamentals (high book-to-market, low PE...)
- ✓ A difference between price and value means that “[...] *there is much inefficiency in the market.*” (p.13)